Shifting Drivers of Stock Returns

Machine learning evidence on time-varying predictability in US equities, 1980–2024

Max Cordier & Robbe Denaux · Master of Business Administration – Financial Analysis · Promoter: Prof. Dr. Chunjie Wang · KU Leuven, Faculty of Economics and Business

1 Research Question

How do top firm-level characteristics identified by machine learning techniques as predictors of stock returns vary over time?

Classical predictors such as value and momentum have weakened substantially in recent decades, while size and liquidity-based signals appear more persistent. Whether the cross-sectional return structure is stable or shifts across market regimes is an open empirical question.

Machine learning is well-suited to tracking this variation: it evaluates large predictor sets simultaneously, captures nonlinear interactions between firm characteristics, and applies disciplined out-of-sample validation across the full cross-section of US equities.

2 Data & Methodology

SAMPLE	FIRMS
1.28M	**12,624**
firm-month observations	*unique US-listed firms*

PERIOD	PREDICTORS
1970–2024	**49**
45 rolling 10-year windows	*from CRSP + Compustat via WRDS*

Rolling-window design. Fixed 10-year training window followed by a 1-year out-of-sample prediction, shifted forward annually. 45 windows in total (predicting 1980–2024).

Three ML models. ElasticNet (regularized linear), Random Forest (tree ensemble), and a feed-forward Neural Network, spanning the interpretability–flexibility trade-off.

Hyperparameter tuning. Per window (EN); per decade (RF, NN). Optimal model complexity may itself change across regimes.

Importance metrics. Absolute coefficient magnitude (EN); permutation importance (RF, NN). An alpha-grid analysis tracks the order in which features enter the model as regularization is progressively relaxed.

Predictive performance evaluation. Out-of-sample R^2 is measured per model per window.

Portfolio evaluation. Equal-weighted decile portfolios sorted on predicted return each month; long-short (D10 – D1) returns used to assess economic significance through Sharpe ratio and decile return monotonicity.

3 Contribution

Extends Gu, Kelly & Xiu (2020) by replacing expanding-window estimation with rolling decades and per-decade hyperparameter tuning.

Documents systematic shifts in predictor importance across four decades of US equity data.

Complements the anomaly-decay literature by analyzing decay and emergence across 49 characteristics simultaneously, rather than one anomaly at a time.

4 Key Results

Out-of-sample R^2. Mean OOS R^2 is negative for all three models — EN: −0.4%, NN: −1.3%, RF: −7.2% — consistent with the broader ML asset pricing literature (Gu et al. report ~0.4% for linear models on an expanding window). Negative OOS R^2 in cross-sectional return prediction is expected and does not preclude strong economic value: **all three models generate strongly positive Sharpe ratios and substantially outperform the equal-weighted benchmark.**

Annualized long-short Sharpe ratios, full sample 1980–2024

2.59	2.12	1.58
ElasticNet	Neural Network	Random Forest

Long-short portfolios from all three models substantially outperform the equal-weighted market over the full sample.

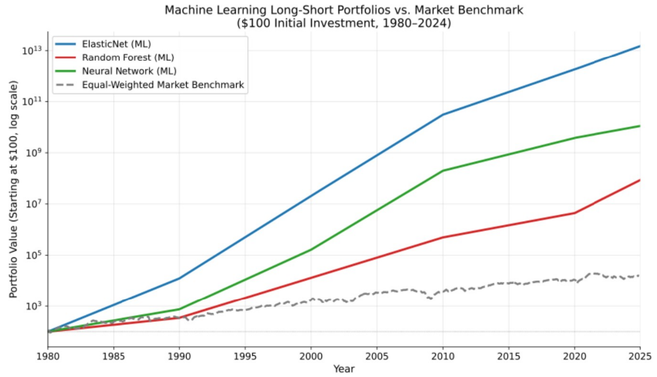


Figure 1 — Machine learning long-short portfolios vs. equal-weighted market benchmark, 1980–2024 ($100 initial investment, log scale).

Realized returns rise monotonically from low to high predicted decile — models genuinely rank stocks, not just identify tail outperformers.

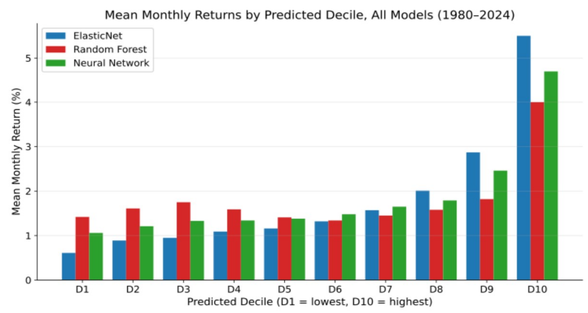


Figure 2 — Mean monthly return by predicted decile across the three models, full sample (1980–2024).

Statistical and economic significance both hold up. ElasticNet produces the cleanest monotonic ranking; all three models substantially outperform the equal-weighted market, with positive long-short returns in every decade from 1980 to 2024.

5 Time-Varying Predictor Importance

Entry order of the 15 most important features as ElasticNet regularization is progressively relaxed, by decade.



Figure 3 — Lower (greener) ranks indicate a feature is selected earlier as a decreases, and is therefore more informative for that decade.

THREE KEY FINDINGS

Size persists. log_mvel1 and mvel1 rank in the top 5 in every decade for all three models — the most robust signal in the cross-section.

Momentum recedes. Dominant in the 1980s–90s; drops out of the top 10 by the 2010s, consistent with post-publication anomaly decay (McLean & Pontiff, 2016).

Value re-emerges. Book-to-market rises from rank 14–15 in the 1980s–90s to rank 10 in the 2010s — the value premium has shifted in expression, not disappeared (Israel et al., 2021).

6 Conclusion

Predictor importance is regime-dependent, not stable. The cross-sectional return structure has evolved meaningfully across four decades: momentum-dominated in the 1980s, liquidity-led in the 2000s, and accounting-driven in the 2010s–2020s. Treating predictor importance as fixed misses these shifts; tracking it dynamically reveals them.

Implication for practice. A quantitative strategy calibrated on a long historical sample may underweight characteristics that have become more important recently and overweight characteristics whose informational content has decayed. Predictor weights should be re-estimated periodically rather than treated as time-invariant.

KEY REFERENCES Gu, S., Kelly, B., & Xiu, D. (2020). Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5). · McLean, R. D., & Pontiff, J. (2016). Does academic research destroy stock return predictability? *Journal of Finance*, 71(1). · Israel, R., Laursen, K., & Richardson, S. (2021). Is (systematic) value investing dead? *Journal of Portfolio Management*, 47(2).